

Challenges and Success Factors for Large-scale Agile Transformations — A Research Proposal and a Pilot Study

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ABSTRACT

Agile methods have become an appealing alternative for large companies striving to improve their performance, even though the methods were originally designed for small, single teams. Our recently performed systematic literature review on large-scale agile transformations revealed that despite the popularity of the topic in the industry, it has received very little research attention, as almost 90% of the selected papers were experience reports. In our literature study, we identified 35 challenges grouped into nine categories, and 29 success factors grouped into eleven categories.

In this research proposal we suggest a survey research to understand these factors with a larger population of companies. The purpose is to validate and deepen our literature study findings. We report the findings from a pilot study performed in the XP2016 Large-scale agile workshop, where we received 19 answers to the survey and several comments from 16 persons participating in the WorldCafe discussion groups. The improved survey will be rolled out to a large number of companies.

CCS Concepts

•Social and professional topics → Project and people management; •Software and its engineering → Agile software development;

Keywords

large-scale agile software development, research proposal

1. INTRODUCTION

Agile methods have become an appealing alternative for large companies striving to improve their performance, even though the methods were originally designed for small and individual teams [1]. Compared to small organizations ideal for agile development, large organizations are characterized by the need for additional coordination. Besides inter-team coordination needs, large organizations also tend to have

additional organizational units that development must interface with, such as human resources, marketing and sales, and product management, which creates extra complexity. Despite these challenges, there seems to be an industry trend towards adopting agile methodologies also in large software development organizations [16, 14, 13, 6]. During recent years, consultants and practitioners have put forward several frameworks for scaling agile, such as Large-scale Scrum (LeSS) [9] and Scaled Agile Framework (SAFe) [10] to help companies scale agile. Currently, the "Agile Scaling Knowledgebase Decision Matrix" [15], lists nine different approaches or frameworks for scaling agile. The timeliness and importance to practitioners is evidenced, e.g., through the fact that recent years have seen a number of books published and courses organized by consultants on scaling agile, and a large number of talks on this topic at agile conferences.

Freudenberg and Sharp [8] asked the industrial practitioners at the XP2010 conference to create a backlog of topics they think should be studied. The practitioners voted "Agile and large projects" as the top burning research question. Moreover, in two recent workshops on large-scale agile development organized in XP2013 and XP2014 conferences, adoption of agile methods was one of the highlighted themes needing more research [5, 6]. Despite the clear industry interest and trend to use agile in large-scale organizations there exists surprisingly little research on the topic. Our recently conducted systematic literature review [4] identified only six research papers and 46 experience reports on adopting agile at large scale. Thus, there exists a clear research need in the area of large-scale agile.

We identified two areas to study on large-scale agile: 1) how to introduce agile in a large organization, and 2) how to scale the agile practices to large-scale settings. Both topics are important for companies. In our study, we decided to concentrate on the first mentioned topic.

Introducing agile methods in large organizations is more difficult than in small ones [7]. The difficulty is partly related to size bringing higher organizational inertia which slows down organizational change [11]. Agile development is not founded on the use of individual tools or practices, but rather on a holistic way of thinking. Adopting agile often requires change to the organizational culture [12].

In the SLR we identified 35 challenges of large-scale agile adoptions grouped into nine categories, and 29 success factors grouped into eleven categories. However, based on the identified papers we could not reliably judge the importance of the factors. To this end, we plan to conduct a survey targeting a larger population of companies.

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Surveys on challenges and success factors for agile projects in general have been conducted, e.g. [3]. However, specifically large-scale agile projects have not been scientifically studied. Non-scientific surveys exist, e.g., the State of Agile Survey conducted annually since 2007. The survey has recently included a few questions on large-scale agile as well, e.g. on scaling methods used and tips for success with scaling. According to the latest survey [16] slightly over half of the almost 4000 respondents had more than 1,000 people in their software organization and 45 % of all the respondents worked in development organizations where the majority of their teams were agile. Of course, the sample of this study is limited to a selected subset of companies and countries (of the respondents to the latest survey 65% were from North America and 21% from Europe). However, this indicates that there seems to exist a large number of companies that have taken or are taking agile into use in large-scale settings. Thus, there is a possibility to perform interesting survey studies as well. In this paper we suggest a survey to study how the challenges and success factors recognized by the systematic literature review are experienced in the companies: which ones they have experienced and which ones they consider the most important.

The rest of the paper is structured as follows: Section 2 describes the SLR the proposed survey study is based on, Section 3 presents the survey research proposal, Section 4 describes the pilot study results, Section 5 presents the survey full-scale roll-out plan and Section 6 concludes the paper.

2. BACKGROUND

Recently, we conducted a systematic literature review (SRL) of industrial large-scale agile transformations [4]. To our knowledge no other systematic literature reviews on the usage or adoption of large-scale agile exist. Here, we summarize the process and findings of our literature review, as our research proposal is based on it.

The goal of the literature review was to study how agile methods have been adopted in large-scale settings. Specifically, we aimed to collect both challenges and success factors that have been reported for large-scale agile transformations. Based on earlier studies discussing large-scale agile software development and their interpretations of agile, we defined large-scale to denote *software development organizations with 50 or more people or at least six teams*.

Our keyword search from IEEEExplore, ACM, Scopus and Web of Knowledge found altogether 1875 papers. After filtering according to our inclusion criteria: agile software development, organizational transformation, large-scale and empirical study, we included 52 publications describing 42 industrial cases presenting the process of taking agile methods into use in large organizations. Of these 52 publications a large majority, 46 papers, were experience reports, while only six were research papers including proper research methods. Thus, despite the high popularity and interest of agile also in large organizations, the research seems to be seriously lagging behind. We identified 35 challenges grouped into nine categories, and 29 success factors, grouped into eleven categories, as shown in Tables 1 and 2.

3. RESEARCH PROPOSAL

3.1 Research Goal and Questions

The goal of the proposed research is to understand the identified challenges and success factors of large-scale agile transformations by targeting a larger population of companies. The purpose of this study is to validate and deepen our findings from the literature review. In the SLR we identified a set of challenges and success factors, but were not able to determine their relative importance. In this paper we report the results of a pilot study that will be followed by a large-scale rollout of the survey.

By the suggested study we aim at answering the following research questions:

- RQ1: Which of the challenges and success factors identified in the systematic literature review have been experienced in the industry at large?
- RQ2: How do the practitioners assess the relative importance of the identified challenges and success factors?
- RQ3: Have the practitioners experienced challenges or success factors not identified in the systematic literature review? Which?

3.2 Pilot Study Design

We designed a pilot survey to test the survey and receive feedback based on which we can create the final survey.

In the instrument, we informed the respondents of our definition of large scale, *software development organizations with 50 or more people or at least six teams*.

The first part of the survey concentrated on the challenges and success factors. The respondents were asked to mark *to which extent* they have *experienced* each of the challenges and success factors during their agile transformation using a five point Likert scale (very much – not at all). Using the same scale, respondents were asked to evaluate the *importance* of each challenge and success factor. Each challenge and success factor was described by one short sentence. For the pilot study we chose from the literature review the 19 most mentioned challenges (originally 35 challenges) and 18 most mentioned success factors (originally 29 success factors) to make the survey shorter to answer.

In the second part of the survey participants filled in background information: whether they came from research, industry or both; the length of their agile experience; the length of their large-scale agile experience; and experience on large-scale agile transformations.

In the third part of the survey we asked the respondents whether they had experienced other challenges or success factors that were not identified in the systematic literature review. If yes, they were asked to briefly explain what they were in open text fields and indicate their importance.

Finally, the pilot questionnaire had an open field for feedback on the survey to help further improve it.

3.3 Pilot Study at XP2016

We piloted the survey with practitioners and researchers in the XP2016 International Workshop on Large Scale Agile Development held in Edinburgh on May 24th, 2016. This full-day workshop had 30 participants.

We had two short sessions on our study. In the first one, during the morning, we briefly presented the background and goals of the research, and asked the participants to fill in the pilot survey. In the second session, in the afternoon,

Table 1: Challenges (modified from [4])

Change resistance General resistance to change Skepticism towards the new way of working Top down mandate creates resistance Management unwilling to change	Hierarchical management and organizational boundaries Middle managers' role in agile unclear Management in waterfall mode Keeping the old bureaucracy Internal silos kept
Lack of investment Lack of coaching Lack of training Too high workload Old commitments kept Challenges in rearranging physical spaces	Requirements engineering challenges High-level req. management largely missing in agile Requirement refinement challenging Creating and estimating user stories hard Gap between long and short term planning
Agile difficult to implement Misunderstanding agile concepts Lack of guidance from literature Agile customized poorly Reverting to the old way of working Excessive enthusiasm	Quality assurance challenges Accommodating non-functional testing Lack of automated testing Requirements ambiguity affects QA
Coordination challenges in multi-team environment Interfacing between teams difficult Autonomous team model challenging Global distribution challenges Achieving technical consistency	Integrating non-development functions in the transformation Other functions unwilling to change Challenges in adjusting to incremental delivery pace Challenges in adjusting product launch activities Rewarding model not teamwork centric
Different approaches emerge in a multi-team environment Interpretation of agile differs between teams Using old and new approaches side by side	

Table 2: Success Factors (modified from [4])

Management support Ensure management support Make management support visible Educate management on agile	Training and coaching Provide training on agile methods Coach teams as they learn by doing
Commitment to change Communicate that change is non-negotiable Show strong commitment	Engaging people Start with agile supporters Include persons with previous agile experience Engage everyone in the organization
Leadership Recognize the importance of change leaders Engage change leaders without baggage of the past	Communication and transparency Communicate the change intensively Make the change transparent Create and communicate positive experiences in the beginning
Choosing and customizing the agile approach Customize the agile approach carefully Conform to a single approach Map to the old way of working to ease adaptation Keep it simple	Mindset and Alignment Concentrate on agile values Arrange social events Cherish agile communities Align the organization
Piloting Start with a pilot to gain acceptance Gather insights from a pilot	Team autonomy Allow teams to self-organize Allow grass roots level empowerment
	Requirements management Recognize the importance of the Product Owner role Invest in learning to refine the requirements

we first discussed the survey and how to improve it further in the WorldCafe [2] format with the participants. We had three groups of altogether 16 participants and 20 minutes discussion within each group led by one of the researchers and documented on flip charts. In the end, we presented the main results of the pilot study to the workshop audience.

In the pilot study we did not limit the respondents to only those having experienced large-scale agile adoption. All workshop participants having some experience from agile adoptions regardless of the size of their organization or whether their background was from research or industry, were invited to answer. That way we could maximize the number of people giving us feedback on the survey.

We received a total of 25 answers, of which 19 were complete. Of the respondents of the completed surveys eight were academic, eight both from the industry and academia, and three from the industry. The respondents had observed between zero and 40 large-scale agile transformations, with a mean of 5.72, and had actively participated in zero to twenty transformations, with the mean being 3.45. 18 respondents had either observed or participated in at least one large-scale agile transformation. With respect to years of experience with large-scale agile, the median was 3-5 years, and the mode over 8 years, only one respondent reported not having large-scale agile experience.

4. PILOT STUDY RESULTS

4.1 Findings

Interestingly, the pilot study results somewhat differed from the systematic literature review findings regarding the most often mentioned challenges and success factors. The top challenges and success factors based on the pilot survey are shown in Table 3.

The top challenges from the literature review, based on the number of cases that mentioned a specific challenge, were: 1) *other functions unwilling to change* (31% cases), 2) *lack of guidance from the literature* (21%), 3) *misunderstanding agile concepts* (19%) and *reverting to the old way of working* (19%). Contrasting this to the pilot study results in Table 3, we can see that the five most often experienced challenges and the five most important challenges reported in the pilot study had two items exactly the same, *challenges in coordinating the work of several teams* and *management in waterfall mode*, while none of the five top items matched the literature top items.

The top success factors based on the number of cases where mentioned in the literature review were: 1) *ensure management support* (29%) and *coach teams as they learn by doing* (29%), and 2) *customize the agile approach carefully* (26%). Regarding success factors, the pilot survey respondents found the top five most important and the top five most experienced to be exactly the same. The top success factor of both categories, *management support*, also matched to the literature findings by being the top success factor also there. However, the other top literature study categories did reach the top five in the pilot study.

A few pilot study respondents suggested new challenges and success factors to be added, but none of them were suggested by more than one person.

4.2 Improvement Suggestions

The most important improvement suggestions that arose

Table 3: Pilot study results

Experienced Challenges	Mean
Challenges in coordinating the work of several teams	3.95
Management in waterfall mode	3.89
Rewarding model not team-work centric	3.79
Achieving technical consistency between teams, e.g. integration, interfaces	3.68
Middle managers' role in agile unclear	3.63
Challenge importance	Mean
Management in waterfall mode	4.26
General resistance towards change	4.06
Challenges in coordinating the work of several teams	3.95
Achieving technical consistency between teams, e.g. integration, interfaces	3.89
Lack of agile coaching	3.74
Experienced success factors	Mean
Management support	4.05
Recognizing the importance of the PO role	3.95
Providing training on agile methods	3.79
Showing strong commitment to the transformation	3.89
Creating and communicating positive experiences in the beginning	3.74
Success factor importance	Mean
Management support	4.68
Recognizing the importance of the PO role	4.42
Providing training on agile methods	4.32
Showing strong commitment to the transformation	4.26
Creating and communicating positive experiences in the beginning	4.26

both in the open field comments and from the WorldCafe discussions were: 1) in the current format the survey takes too long to answer (now 10 min or less) and 2) having the two scales, the *importance* of each challenge or success factor and *to which extent* each of them have been *experienced* requires a lot of thinking and might make answering confusing.

Our respondents hoped for a bit more structure to the survey, which they now experienced as "too flat". Some said that regarding many of the challenges and success factors they had given the highest points as they had seen them important. However, they would have liked the answerers to choose only the most important ones. For example, by first choosing the ten most important success factors they had seen, and then giving order to the five most important.

In addition, we received a large number of suggestions on additional questions to add to the survey, especially regarding the background of a specific transformation that could then be used to compare with and explain the differences between transformations and answers given. Interesting context information on specific transformations would be the scaling framework applied (e.g. SAFe, LeSS, DAD), whether the transformation was successful or not, who initiated the transformation (e.g. was it bottom up or top down), the phase or maturity level of the transformation while answering the survey, whether the transformation is made on a company or a project level, and whether the transformation is made in a product development program or a in customer specific project. Even though all this information would provide very interesting additions to the survey, the challenge would be to require the respondents to choose and think about only one specific transformation while answering the survey, or to fill in the survey several times regarding each

transformation separately.

Regarding the the pilot study we asked the respondents to think about all large-scale agile transformations they had experienced to make answering the survey as easy as possible, and to receive answers that are based on all their experience. Most of the respondents reported having experienced or observed several large-scale agile transformations. Four persons had observed and / or participated in one transformation, whereas most had observed / participated between 2 to 10 transformations, the largest number of transformations participated in being 20. The respondents hoped to see background questions already in the beginning of the survey to make sure that a respondent knows that context is asked and to clarify which context he or she should think about.

Other important questions that came up in the discussions were: who in an organization should answer the survey, and how many persons from one organization should answer. It was also suggested that the survey could be somewhat different when targeted to different organizational levels. Finally, one interesting suggestion was that this survey could be used as a coaching tool in large-scale agile transformations and the whole personnel could be asked to answer to the survey.

5. ROLL-OUT

We intend to improve the survey based on the pilot study results and feedback. For the finalized survey, we plan to get as large as possible set of practitioners of large-scale agile development to participate in. To reach this aim, we plan to contact agile practitioners through various channels, including LinkedIn, the Agile Alliance, the Scrum Alliance, as well as consultants of large-scale agile.

In the instrument, we ask the respondents who have been involved in a large-scale agile adoption to participate in the survey. Finally, we intend to share our survey results with our respondents and publish them as a journal article.

6. CONCLUSIONS

In this paper, we presented the results of a recent SLR on large-scale agile transformations. In the SLR, we identified a set of challenges and success factors. To validate and deepen our findings from the literature review and to understand the completeness of the identified factors, we plan to conduct a large-scale survey aimed at practitioners of large-scale agile development. Finally, we presented the results of a pilot survey conducted at the International Workshop on Large Scale Agile Development at XP2016 in Edinburgh. The final survey will be improved based on the comments and experiences received.

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8. REFERENCES

- [1] B. Boehm and R. Turner. Management challenges to implementing agile processes in traditional development organizations. *Software, IEEE*, 22(5):30–39, 2005.
- [2] J. Brown. *The World Caf: Shaping Our Futures Through Conversations That Matter (Large Print 16pt)*. ReadHowYouWant. com, 2010.
- [3] T. Chow and D.-B. Cao. A survey study of critical success factors in agile software projects. *J. Syst. Softw.*, 81(6):961–971, June 2008.
- [4] K. Dikert, M. Paasivaara, and C. Lassenius. Challenges and success factors for large-scale agile transformations: A systematic literature review. *Journal of Systems and Software*, 119:87–108, September 2016.
- [5] T. Dingsøy and N. Moe. Research challenges in large-scale agile software development. *SIGSOFT Softw. Eng. Notes*, 38(5):38–39, Aug. 2013.
- [6] T. Dingsøy and N. Moe. Towards principles of large-scale agile development. In T. Dingsøy, N. Moe, R. Tonelli, S. Counsell, C. Gencel, and K. Petersen, editors, *Agile Methods. Large-Scale Development, Refactoring, Testing, and Estimation*, volume 199 of *Lecture Notes in Business Information Processing*, pages 1–8. Springer International Publishing, 2014.
- [7] T. Dybå and T. Dingsøy. Empirical studies of agile software development: A systematic review. *Information and Software Technology*, 50(9-10):833–859, 2008.
- [8] S. Freudenberg and H. Sharp. The top 10 burning research questions from practitioners. *Software, IEEE*, 27(5):8–9, Sept 2010.
- [9] C. Larman and B. Vodde. *Practices for Scaling Lean & Agile Development: Large, Multisite, and Offshore Product Development with Large-Scale Scrum*. Addison-Wesley Professional, 1st edition, 2010.
- [10] D. Leffingwell. *Scaling Software Agility: Best Practices for Large Enterprises (The Agile Software Development Series)*. Addison-Wesley Professional, 2007.
- [11] J. A. Livermore. Factors that significantly impact the implementation of an agile software development methodology. *Journal of Software*, 3(4):31–36, 2008.
- [12] S. C. Misra, V. Kumar, and U. Kumar. Identifying some critical changes required in adopting agile practices in traditional software development projects. *International Journal of Quality and Reliability Management*, 27(4):451–474, 2010.
- [13] M. Paasivaara, B. Behm, C. Lassenius, and M. Hallikainen. Towards rapid releases in large-scale xaas development at ericsson: A case study. In *Global Software Engineering (ICGSE), 2014 IEEE 9th International Conference on*, pages 16–25, Aug 2014.
- [14] M. Paasivaara, C. Lassenius, V. Heikkilä, K. Dikert, and C. Engblom. Integrating global sites into the lean and agile transformation at ericsson. In *Global Software Engineering (ICGSE), 2013 IEEE 8th International Conference on*, pages 134–143, Aug 2013.
- [15] S. Spearman and R. Dolman. Agile scaling knowledgebase decision matrix. <http://www.agilescaling.org/ask-matrix.html>, February 2016.
- [16] VersionOne, Inc. 10th annual "state of agile development" survey. <http://www.versionone.com/pdf/10th-Annual-State-of-Agile-Development-Survey.pdf>, 2015.